Cropify: An Application for Plant Disease Detection

Submitted in partial fulfilment of the requirements of the degree of

BACHELOR OF COMPUTER ENGINEERING

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CERTIFICATE

This is to certify that the project entitled "**Cropify: An Application for Plant Disease Detection**" is a bonafide work of **Vansh J Jain (20102156), Ghanshyam Kachhia (20102026), Yash Kanade (20102043), Vansh V Jain (21202005)** submitted to the University of Mumbai in partial fulfilment of the requirement for the award of the degree of **Bachelor of Engineering** in **Computer Engineering.**

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Project Report Approval for B.E.

This project report entitled **"Cropify: An Application for Plant Disease Detection"** by **Vansh J Jain, Ghanshyam Kachhia, Yash Kanade, Vansh V Jain** is approved for the degree of **Bachelor of Engineering** in **Computer Engineering**, **2023-2024**.

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Abstract

In order to guarantee global food security and economic stability, agriculture is essential. Crop diseases, however, pose a serious danger to crop productivity and quality and are just one of the many difficulties the agricultural industry must overcome. To lessen these dangers, early detection and prompt intervention are necessary. A cutting-edge crop disease prediction application for android to help farmers and agricultural stakeholders efficiently identify and manage crop diseases is developed.

In order to deliver precise and immediate detection of crop illnesses, the Crop Disease Prediction Application makes use of the power of Convolutional Neural Network. With the help of the application, farmers can take pictures of infected crops on their Android handsets, which are then analysed using convolutional neural network. By analysing the photos and comparing them to a huge database of crop disease patterns, the proposed system enables quick diagnosis of the disease impacting potato, tomato, apple and corn crops. The accuracy obtained is 98% for potato, 95% for tomato, 91% for apple and 92% for corn respectively.

Keywords: CNN (Convolutional Neural Network), Deep Learning, Image Recognition, Disease Prediction.

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ABBREVATION

CNN	Convolution Neural Network
DFD	Data Flow Diagram
DL	Deep Learning
ML	Machine Learning
VGG	Visual Geometry Group

CHAPTER 1

Introduction

In an era where global food security was of paramount concern, the development of innovative agricultural technologies was deemed essential to combat the threats posed by crop diseases. The project was focused on the prediction of crop diseases using deep learning techniques, with an emphasis on analyzing images depicting various classes of diseases. Crop diseases had long been a persistent challenge, leading to significant reductions in yield and quality. Traditional methods of disease detection, typically relying on manual visual inspection by trained experts, were not only labor-intensive but also prone to human error, making them far from ideal for large-scale agriculture. The project's primary objective was to revolutionize the field by harnessing the power of deep learning to enable rapid, accurate, and automated crop disease detection.

In recent years, image processing and machine learning techniques had been used to develop automated crop disease detection systems. These systems could be used to rapidly and accurately detect and identify crop diseases from images of plant leaves, stems, and fruits. The project proposed to develop a crop disease prediction system using images of different classes of diseases. The system would be based on a Convolutional Neural Network model, which would be trained on a large dataset of labeled images of healthy and diseased crops. The system would first preprocess the input images to remove noise and improve the quality of the images. Then, it would extract features from the images that were relevant for disease detection. Finally, the system would use the

extracted features to classify the images into healthy and diseased categories. The system would be evaluated on a held-out test set of labeled images. The evaluation metrics would include accuracy, precision, recall, and F1 score. The system was expected to be able to accurately detect and identify a wide range of crop diseases. It was also expected to be robust to variations in lighting, background, and image quality.

The project unfolded through several key phases, starting with the critical step of data collection and preprocessing. To train a robust and reliable deep learning model, an extensive and diverse dataset of labeled images was amassed, encompassing a wide spectrum of both healthy and diseased crops. The images of crops that had been used to train the deep learning model were potato and tomato. These images underwent meticulous preprocessing to eliminate noise and enhance image quality, ensuring that the model learned from pristine data. The subsequent phase entailed feature extraction, in which convolutional neural networks (CNNs) took center stage. These networks extracted pertinent features from the preprocessed images, allowing the model to discern and differentiate between healthy and diseased crops. This feature extraction process was vital in enabling the model to identify the unique characteristics and patterns that distinguished one class of disease from another.

For farmers, the system could be seamlessly integrated into a user-friendly mobile app or web service, providing them with a valuable tool for early disease detection in their fields. This empowerment allowed farmers to take timely and informed actions, mitigating crop losses and promoting higher yields. On a broader scale, the system could be employed for crop disease monitoring, serving as a valuable resource for agricultural scientists and policymakers. By systematically tracking the spread of crop diseases and analyzing patterns, this system equipped decision-makers with the data necessary to develop effective management strategies and safeguard agricultural resources.

In conclusion, the Convolutional Neural Network-based crop disease prediction project, with a focus on analyzing images of different classes of diseases, represented a remarkable convergence of cutting-edge technology and agricultural needs. By automating disease detection, this project addressed a longstanding challenge in agriculture, bridging the gap between traditional farming practices and the boundless possibilities offered by technological innovation.

CHAPTER 2

Literature Survey

Md. Khalid Rayhan Asif, Md. Asfaqur Rahman, Most. Hasna Hena [1], emphasized on the fact that potatoes are a well-known vegetable to all of us. If the other countries are taken into consideration, it can be easily concluded that potatoes are the number one vegetable all over the world, which has been increasingly claimed by many Agricultural departments. The proposed model will strongly identify and detect diseases of potato leaf stand on image processing methods in this research paper.

Quan Huu Cap, Hiroyuki Uga, Satoshi Kagiwada, Hitoshi Iyatomi [7], contains many applications for the automated diagnosis of plant diseases that have been developed based on the success of deep learning techniques. However, these applications often suffer from overfitting, and the diagnostic performance is drastically decreased when used on test data sets from new environments. In this article, we propose LeafGAN, a novel image-to-image translation system with own its attention mechanism. LeafGAN generates a wide variety of diseased images via transformation from healthy images, as a data augmentation tool for improving the performance of plant disease diagnosis. Due to its own attention mechanism, the model can transform only relevant areas from images with a variety of backgrounds, thus enriching the versatility of the training images. A dataset containing images of cucumber leaf

images was used. The average accuracy obtained was 78.7 %. Garima Shrestha, Deepsikha, Majolica Das, Naiwrita Dey [3], mentions the CNN based method for plant disease detection has been proposed here. A total of 15 cases have been fed to the model, out of which 12 cases are of diseased plant leaves namely, Bell Pepper Bacterial Spot, Potato Early Blight, Potato Late Blight, Tomato Target Spot. Laha Ale, Alaa Sheta, Longzhuang Li, Ye Wang, Ning Zhang [8], Deep learning is a promising approach for finegrained disease severity classification for smart agriculture, as it avoids the labor-intensive feature engineering and segmentation based threshold. In this work, we first propose a Densely Connected Convolutional Networks (DenseNet) based transfer learning method to detect the plant diseases, which expects to run on edge servers with augmented computing resources. Then, we propose a lightweight Deep Neural Networks (DNN) approach that can run on Internet of Things (IoT) devices with constrained resources.

	Author	**Paper Name**	**Abstract**	**Research Gap**
[1]	Md. Khalid Rayhan Asif, Md. Asfaqur Rahman, Most. Hasna Hena	CNN based Disease Detection Approach on Potato Leaves	Potatoes are a well-known vegetable to all of us. If the other countries are taken into consideration, it can be easily concluded that potatoes are the number one vegetable all over the world, which has been increasingly claimed by many Agricultural departments. The proposed model will strongly identify and detect diseases of potato leaf stand on image processing methods in this research paper	Data diversity, Model generalization and Real time detection is lacking in this model. It could also be a less complex to make it easier for the farmers to understand

[2]	Sk Mahmudul Hassan, Arnab Kumar Maji	Plant Disease Identificati on Using a Novel Convolutio nal Neural Network	The timely identification of plant diseases prevents the negative impact on crops. Convolutional neural network, particularly deep learning is used widely in machine vision and pattern recognition task. In this paper, they have proposed a novel deep learning model based on the inception layer and residual connection.	There could be an improvement in accuracy of ResNet50 model. There should also be a improvement in the quality of the data. The model also should be able to integrate with the agriculture systems.
[3]	Sunil CK, Jaidhar CD and Nagamma Patil	Cardamom Plant Disease Detection Approach Using EfficientNetV2	In this study, U 2 -Net was used to remove the unwanted background of an input image by selecting multiscale features. This work proposes a cardamom plant disease detection approach using the EfficientNetV2 model. A comprehensive set of experiments was carried out to ascertain the performance of the proposed approach and compare it with other models such as EfficientNet and Convolutional Neural Network (CNN).	This model lacks generalization across plant species. It is necessary to check how well the model will perform on other plant species. The model also need to be optimized to perform in real time. The model also needs to be studied how well it will perform in real world.
[4]	S.Yegneshwar Yadhav, T.Senthilkumar	Plant Disease Detection and Classification	This paper deals with the optimized real time detection of diseases that affect the plant and	There is lack of high quality and diverse dataset. The model is

	, S.Jayanthy, J.Judeson Antony Kovilpillai	using CNN Model with Optimized Activation Function	the area affected using Convolutional Neural Networks (CNN) algorithms so that appropriate fertilizers can be used to prevent further damage to plants from pathogenic viruses. The activation function is the core of the CNN model as it incorporates the non - linearity to have an authentic artificial intelligence system for classification. ReLu is one among the best activation functions, but has a disadvantage that the derivative of the function is zero for negative values and leads to neuronal necrosis. New mathematical activation function is developed and compared with existing activation functions to improve the accuracy and performance of the system on a TensorFlow framework.	also not robust and adaptable. The model needs to be fine tuned as there is lack of data. Multiple data sources should be used.
[5]	Amrita S.Tulshan, Nataasha Raul	Plant Leaf Disease Detection using Machine	According to past studies, 42% of agricultural production is in loss and that too only because of the increasing rate of loss due to	There is a lack of data diversity and quality. Effective features should be selected to

		Learning	plant leaf diseases. To overcome this major issue, this plant leaf disease detection technique can be applied to detect a disease from the input images. This process involved steps like image pre-processing, image segmentation, feature extraction. It also presents other information regarding a plant leaf disease that is Affected Area, Disease Name, Total Accuracy, Sensitivity and Elapsed Time.	ensure performance of the model. The data should be augmented and synthetic data should be generated to address the data scarcity issue.
[6]	Quan Huu Cap, Hiroyuki Uga, Satoshi Kagiwada, Hitoshi Iyatomi	LeafGAN: An Effective Data Augmentation Method for Practical Plant Disease Diagnosis	Many applications for the automated diagnosis of plant disease have been developed based on the success of deep learning techniques. However, these applications often suffer from overfitting, and the diagnostic performance is drastically decreased when used on test data sets from new environments. In this article, we propose LeafGAN, a novel image-to-image translation system with own attention mechanism. LeafGAN generates a wide variety of	There is lack of quality of augmented data. Augmentation techniques used must be evaluated. The complexity could be decreased to make users understand why model makes predictions. The model needs to be studied to be applied in practical applications.

			diseased images via transformation from healthy images, as a data augmentation tool for improving the performance of plant disease diagnosis. Due to its own attention mechanism, our model can transform only relevant areas from images with a variety of backgrounds, thus enriching the versatility of the training images.	
[7]	Laha Ale, Alaa Sheta, Longzhuang Li, Ye Wang, Ning Zhang	Deep Learning based Plant Disease Detection for Smart Agriculture	Deep learning is a promising approach for fine- grained disease severity classification for smart agriculture, as it avoids the labor-intensive feature engineering and segmentation-based threshold. In this work, we first propose a Densely Connected Convolutional Networks (DenseNet) based transfer learning method to detect the plant diseases, which expects to run on edge servers with augmented computing resources. Then, we propose a lightweight Deep Neural Networks (DNN) approach that	There needs to be more investigation on how to overcome the practical challenges associated with deploying deep learning-based disease detection systems on the farm, including issues related to hardware, connectivity, and user interfaces. There should also be research on the privacy and security considerations when collecting and transmitting data for disease detection in smart agriculture.

			can run on Internet of Things (IoT) devices with constrained resources.	Protecting sensitive agricultural data is important.
[8]	Divyansh Tiwari, Mritunjay Ashish, Nitish Gangwar, Abhishek Sharma, Suhanshu Patel, Dr. Suyash Bhardwaj	Potato Leaf Diseases Detection Using Deep Learning	With the enhancement in agricultural technology and the use of artificial intelligence in diagnosing plant diseases, it becomes important to make pertinent research to sustainable agricultural development. In this paper, a model is presented that uses pre-trained models like VGG19 for fine-tuning(transfer learning) to extract the relevant features from the dataset. Then, with the help of multiple classifiers results were perceived among which logistic regression outperformed others by a substantial margin of classification accuracy obtaining 97.8% over the test dataset.	The interpretability and explainability of deep learning models in potato leaf disease detection should be addressed. Farmers and stakeholders may need to understand why a specific diagnosis was made. The potential of disease early detection should be explored and implemented. The ability of deep learning models to generalize across different potato varieties, geographic locations, and environmental conditions must be tested. Models that perform well in one context may not be effective in others.
[9]	Marwan Adnan Jasim, Jamal Mustafa AL-	Plant Leaf Diseases Detection and	Agricultural products are the primary need for every country. If plants are infected by	The data used in this model is unbalanced as it does not cover

	Tuwaijari	Classification Using Image Processing and Deep Learning Techniques	diseases, this impacts the country's agricultural production and its economic resources. This paper presents a system that is used to classify and detect plant leaf diseases using deep learning techniques. The used images were obtained from (Plant Village dataset) website. In our work, we have taken specific types of plants; include tomatoes, pepper, and potatoes, as they are the most common types of plants in the world and in Iraq in particular. This Data Set contains 20636 images of plants and their diseases. In our proposed system, we used the convolutional neural network (CNN), through which plant leaf diseases are classified, 15 classes were classified, including 12 classes for diseases of different plants that were detected, such as bacteria, fungi, etc., and 3 classes for healthy leaves.	different environment conditions. Different optimizers and learning rates should be applied to improve the efficiency of the model. Different techniques can also be used to improve accuracy of the model.
[10]	Huiqun Hong, Jinfa Lin,	Tomato Disease	Tomato is widely cultivated in China as a kind of fruit or	The parameters of the dataset could be

	Fenghua Huang	Detection and Classification by Deep Learning Restructured Deep Residual Dense Network	vegetable. In the whole growth cycle of tomatoes, there are many types of tomato diseases and pests, therefore, the detection and diagnosis of these diseases are very necessary. In this research, transfer learning is used to reduce the size of the training data, the time and the computational costs when building deep learning. And 9 types of disease leaves including healthy tomato leaves are classified. Five deep network structures of Resnet50, Xception, MobileNet, ShuffleNet, Densenet121_Xception were applied to perform the feature extraction. Those network structures with different learning rates were compared in the experiment. Through comparison in this paper, which provides model support for the further development of an intelligent tomato disease diagnosis system based on smart phones and other mobile terminals, and is of great	increased for this model. The accuracy of deep network structures apart from Densenet_Xception could be improved. The practical use of model needs to be studied.

			significance for assisting tomato pest control decision-making.	
[11]	Changjian Zhou, Sihan Zhou, Jinge Xing, and Jia Song	Tomato Leaf Disease Identification by Restructured Deep Residual Dense Network	In this paper, a restructured residual dense network was proposed for tomato leaf disease identification; this hybrid deep learning model combines the advantages of deep residual networks and dense networks, which can reduce the number of training process parameters to improve calculation accuracy as well as enhance the flow of information and gradients. The original RDN model was first used in image super resolution, so we need to restructure the network architecture for classification tasks through adjusted input image features and hyper parameters. Experimental results show that this model can achieve a top-1 average identification accuracy of 95% on the Tomato test dataset in AI Challenger 2018 datasets, which verifies its satisfactory performance. The restructured	The model needs to be tested in different environment conditions. It is also necessary to check how well the model will perform on other plants and crops. There is also a lack of implementation for practical use.

			residual dense network model can obtain significant improvements over most of the state-of-the-art models in crop leaf identification, as well as requiring less computation to achieve high performance.	

Table 2.1: Literature Survey Table

CHAPTER 3

Limitation of Existing System

Like any predictive system, crop disease prediction models have some drawbacks that may reduce its precision and efficiency. To use these models successfully and make wise agricultural decisions, it is crucial to be aware of these limitations. The following list of typical model limitations for crop disease prediction:

- **Data Quantity and Quality:** The quality and quantity of data used for training has a significant impact on the accuracy and reliability of crop disease prediction models. Inaccurate predictions may result from incomplete, biased, or outdated training data. Additionally, there may be restrictions on data gathering in specific areas or for particular crops, which could impact model performance.

- **Resource Intensity:** For training and inference, some crop disease prediction algorithms require significant computational resources. Farmers or areas with limited access to technology or internet connectivity may find this to be a restriction.

- **Model overfitting:** When a model fits training data too closely, noise in the data is captured instead of underlying patterns. Poor generalisation to fresh, unforeseen facts may come from this.

- **Imperfect Accuracy:** No system is entirely error-free. Crop disease prediction systems may produce false positives, causing unnecessary alarm, or false negatives, failing to detect actual disease cases, which can lead to crop loss if not addressed promptly.

- **Limited Coverage of Disease Spectrum:** Many crop disease prediction models focus on a subset of prevalent diseases or common pathogens, potentially overlooking emerging or rare diseases that may pose a threat to crops. This limited coverage can leave farmers vulnerable to unexpected outbreaks and hinder proactive disease management strategies.

- **Accessibility Barriers:** Limited accessibility to technology and digital infrastructure in rural or developing regions can impede farmers ability to access and benefit from crop disease prediction systems. This digital divide exacerbates disparities in agricultural productivity and resilience to disease outbreaks.

CHAPTER 4

Problem Statement, Objectives and Scope

4.1 Problem Statement

Developing an accurate and accessible crop disease prediction system to mitigate yield losses and ensure food security in the face of evolving agricultural challenges.

4.2 Objectives

- To detect diseases in crops at their earliest stages, often before visible symptoms appear, to allow for timely intervention.
- Promote data-driven decision-making in agriculture by utilizing technology and machine learning to analyze and predict disease outbreaks.

4.3 Scope

- Create a mobile application that allows farmers to upload images of their plants to predict the diseases present in plants.
- The application uses image recognition and machine learning to identify the diseases in plants, tomato and potato.
- Develop a deep learning model that can identify plant diseases from leaf images. The model is trained on a large dataset of labeled images of diseased and healthy leaves

CHAPTER 5

Proposed System

5.1 Architecture Diagram

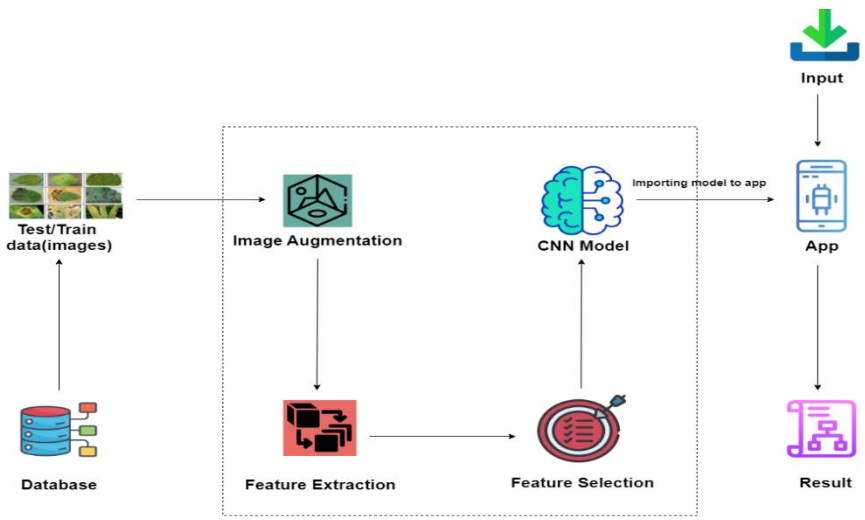

Fig 5.1: Architecture Diagram

In the architecture diagram, the data from the dataset is passed through an image augmentation step to obtain different types of images. After that, this augmented data is passed through a feature extraction step so that different features are extracted. After that, some features are selected from

the feature extraction data. After that, a CNN model is applied to this data. The CNN model is then converted into a h5 file and imported into an application that predicts the result whenever an input image is provided.

5.2 Design Details

5.2.1 DFD Diagram

DFD or Data Flow Diagram shows the flow of information in a system.

i. DFD level 0

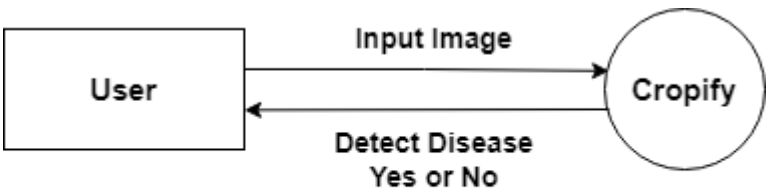

Fig 5.2.1.i: Data Flow Diagram Level 0

The data flow diagram level 0 for crop disease prediction shows that:

User: The user inputs an image of a crop leaf.

Cropify: The Cropify system takes the image input from the user and predicts whether the leaf is healthy or diseased.

Prediction: The Cropify system outputs a prediction of whether the leaf is healthy or diseased to the user.

The data flow between the user and Cropify is shown as a single arrow, indicating that the system takes the image input from the user and produces a prediction output.

ii. DFD level 1

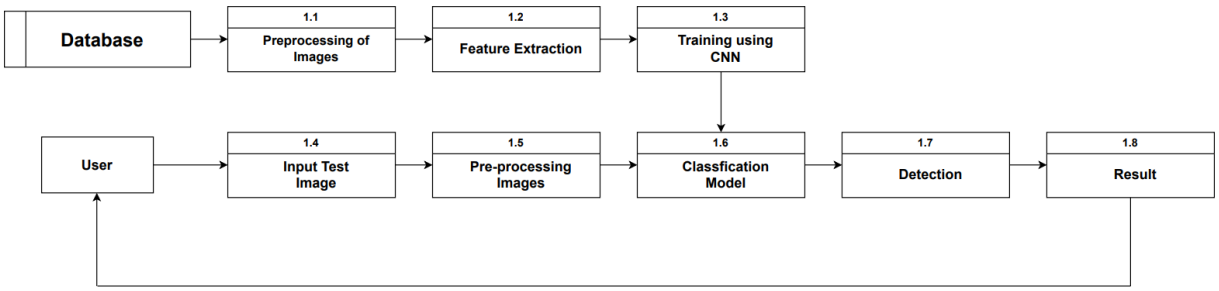

Fig 5.2.1.ii: Data Flow Diagram Level 1

The level 1 data flow diagram for crop disease prediction shows the following steps:

User: The user provides an input test image of a crop leaf.

Pre-processing Images: The input image is pre-processed to improve its quality and consistency. This may involve steps such as resizing, denoising, and normalization.

Feature Extraction: Features are extracted from the pre-processed image. These features may include color, texture, and shape features.

Model: The extracted features are passed to a machine learning model to predict the presence or absence of a crop disease.

Detection: The model outputs a prediction, which is then used to detect the presence or absence of a crop disease in the input image.

Result: The result of the detection is displayed to the user.

5.2.2 Activity Diagram

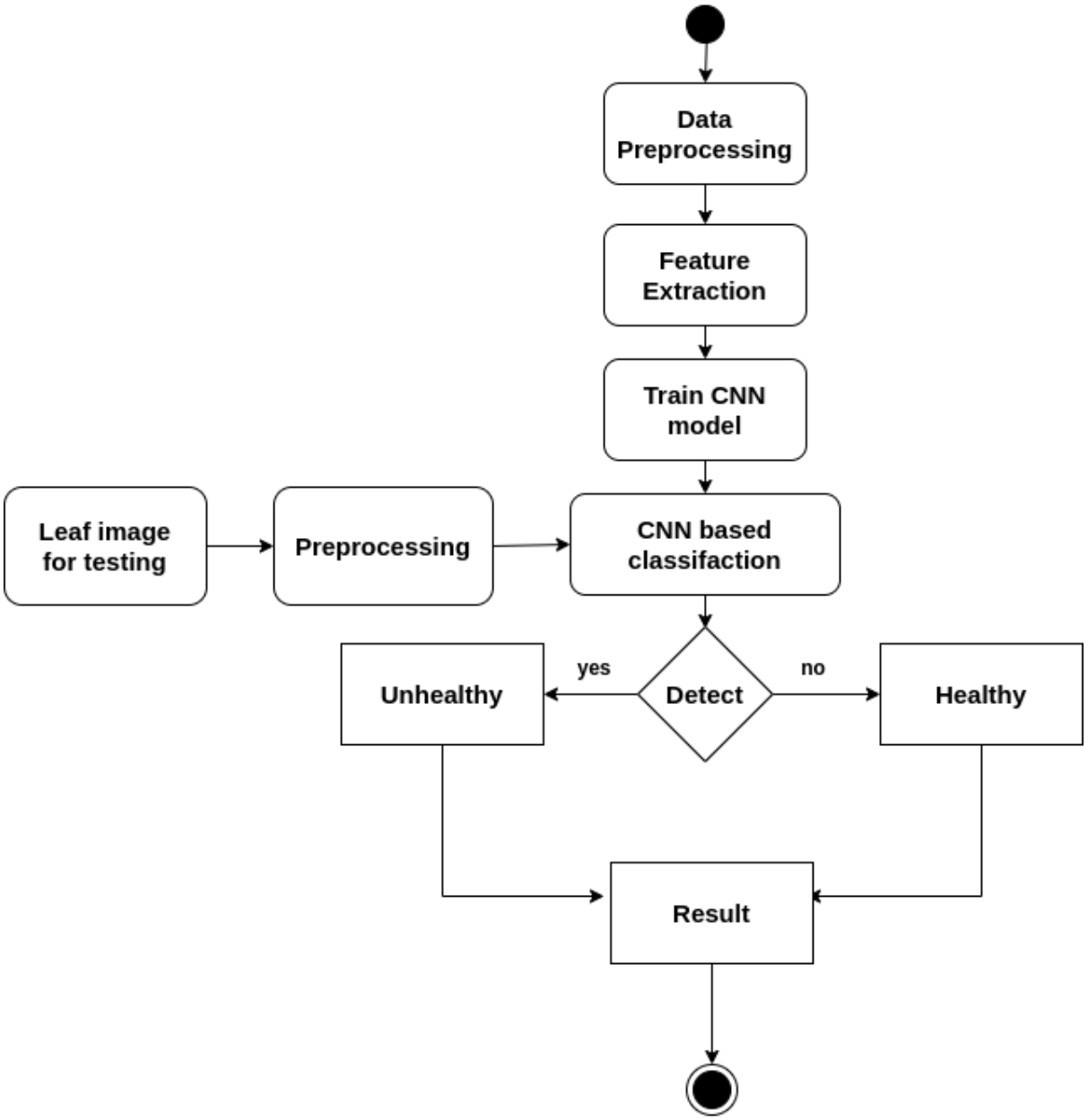

Fig 5.2.2: Activity Diagram

The leaf image testing diagram shows the steps involved in using a convolutional neural network (CNN) to classify a leaf image as healthy or unhealthy. The first step is to preprocess the image by resizing it to a consistent size and normalizing the pixel values. This helps to ensure that the CNN can learn general features of leaves, rather than being biased towards specific leaf types or sizes. Once the image has been preprocessed, it is passed to the CNN. The

CNN will extract features from the image, such as the shape, color, and texture of the leaf. These features are then used to train the CNN to predict whether the leaf is healthy or unhealthy. Once the CNN has been trained, it can be used to classify new leaf images. To do this, the image is preprocessed and then passed to the CNN. The CNN will then predict whether the leaf is healthy or unhealthy.

5.2.3 Sequence Diagram

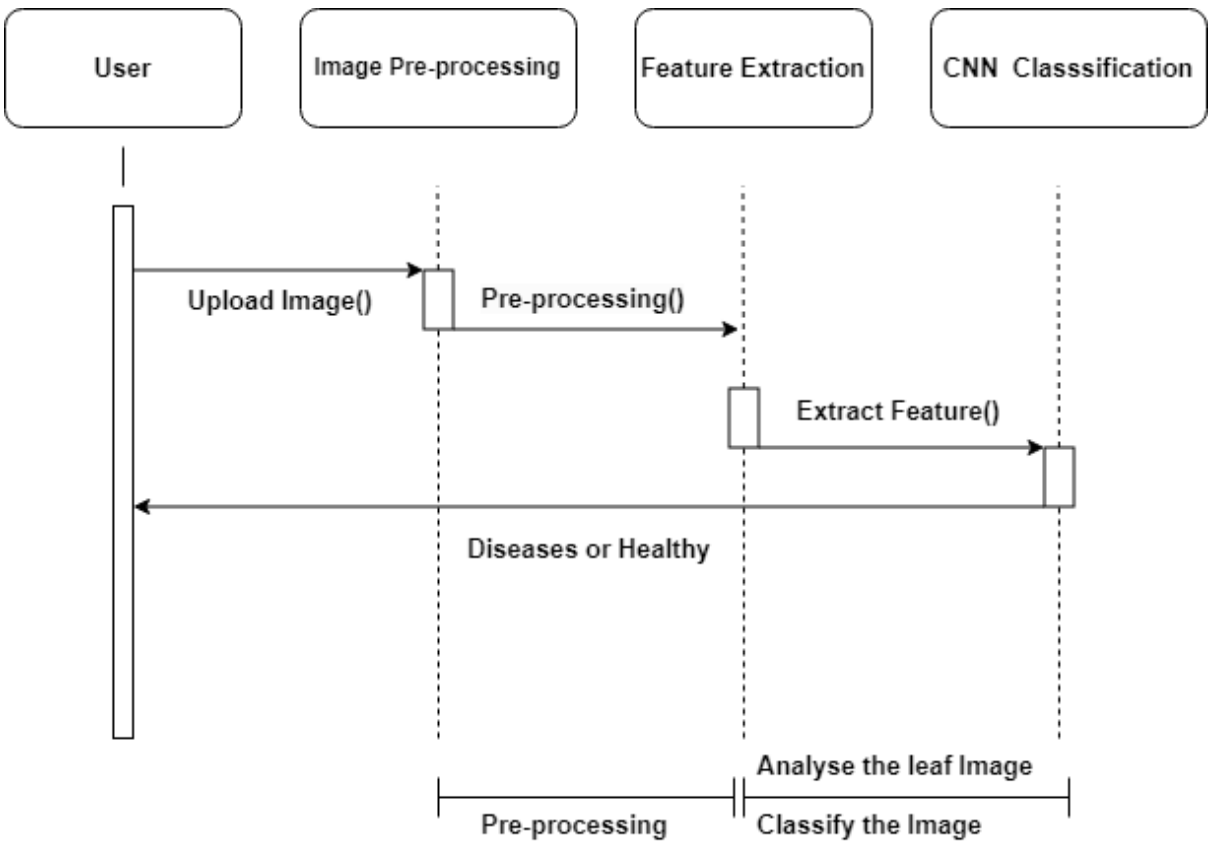

Fig 5.2.3: Sequence Diagram

The sequence diagram shows the steps involved in analyzing a leaf image using a convolutional neural network (CNN).

Upload Image(): The user uploads a leaf image to the system.

Preprocessing(): The system preprocesses the image by resizing it to a consistent size and normalizing the pixel values.

Extract Feature(): The system extracts features from the image, such as the shape, color, and texture of the leaf.

CNN Classification(): The system uses a CNN to classify the image as healthy or unhealthy.

Analyse the leaf Image: The system displays the results of the analysis to the user, including the classification of the image and any other relevant information.

5.3 Methodology

Define Objectives and Scope:

The aim of the crop disease prediction application should be clearly defined. The types of crops and diseases the application is going to cover should be mentioned. The final goal of the application should be to achieve a high level accuracy.

1. **Data Collection:**

- Gather a comprehensive dataset of images and information about crop diseases. The data can be collected from various sources, such as agricultural research institutions, universities, and online databases.
- Ensure the dataset is diverse, representing different types of crops and diseases, and it should be labeled correctly.

2. **Data Preprocessing:**

- Clean the data by removing duplicates, low-quality images, and irrelevant information.
- Split the data into training, validation, and testing sets.
- Augment the data to increase its diversity and improve the model's ability to generalize.

3. **Select a Machine Learning Model:**

- Choose a suitable machine learning model for image classification. Convolutional Neural Networks (CNNs) are a popular choice for this type of task.
- Pre-trained models such as ResNet, VGG, or Inception can be fine-tuned for your specific task.

4. Model Training:

- Train the selected model on the training data.
- Monitor the training process, adjusting hyperparameters as needed.
- Use transfer learning if possible to leverage pre-trained models and speed up training.

5. Validation and Evaluation:

- Validate the model's performance on the validation dataset.
- Use metrics like accuracy, precision, recall, and F1-score to evaluate its performance.
- Iterate and fine-tune the model as necessary.

6. Testing and Fine-Tuning:

- Test the model on the testing dataset to get an accurate assessment of its real-world performance.
- Make necessary adjustments to the model based on the test results.

7. User Interface Design:

- Design an intuitive and user-friendly interface for the application.
- Include features like image uploading, real-time camera input, and user-friendly disease identification reports.

8. App Development:

- Develop the front-end and back-end of the application.
- Integrate the trained machine learning model into the application.
- Ensure the application is responsive and works well on various devices

CHAPTER 6

Experimental Setup

6.1 Technology Stack

6.1.1 Software Requirements: -

- **Programming Language:** Python, Java, Kotlin.
- **Machine Learning Library:** Scikit-Learn
- **Data Visualization Tool:** Matplotlib,Numpy,Pandas
- **Integrated Development Environment:** Jupyter Notebook

6.1.2 Hardware Requirements: -

- **CPU:** A modern multicore CPU is sufficient for training small to medium-sized datasets. However, for large datasets or complex models, a high-end CPU with more cores can provide faster training times.
- **GPU:** GPUs are recommended because they can perform matrix operations much faster than CPUs.
- **Memory (RAM):** Sufficient memory is required to store the dataset and the model parameters. The minimum recommended RAM is 8 GB, but for larger datasets and models, 16 GB or more is recommended.

CHAPTER 7

Implementation and Results

7.1 Implementation

Farmers can use the application to snap images of infected crops on their Android devices, which are then evaluated using a convolutional neural network. The technique allows for identification of the specific disease affecting potatoes, tomatoes, corn and apples by processing the images and comparing them to a massive database of crop disease patterns.

This code sets a click listener on an ImageView. It verifies whether the app has been granted the CAMERA permission. If the permission is granted, it initiates an intent to open the camera for image capture. If the permission is not granted, it requests the CAMERA permission from the user. This code ensures runtime permission handling for accessing the device's camera.

```
imageview1.setOnClickListener(new View.OnClickListener() {
            @RequiresApi(api = Build.VERSION_CODES.M)
            @Override
            public void onClick(View view) {
                if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.M) {
                    if (checkSelfPermission(Manifest.permission.CAMERA) ==
PackageManager.PERMISSION_GRANTED){
                        Intent cameraIntent = new
Intent(MediaStore.ACTION_IMAGE_CAPTURE);
                        startActivityForResult(cameraIntent, Camera_code);
                    }else {
                        requestPermissions(new
String[]{Manifest.permission.CAMERA},100);
                    }
                }
```

```
            }
        });
```

This code sets a click listener on a TextView. Upon clicking, it creates an intent to pick an image from the device's gallery. The intent is initialized with the action Intent.ACTION_PICK, indicating selection from data. It specifies the data source as the external content URI for images using MediaStore.Images.Media.EXTERNAL_CONTENT_URI. Then, it starts the activity for result with the intent, expecting a response denoted by GALLERY_REQ_CODE. This code enables the user to select an image from the device's gallery and handle the selected image accordingly within the app.

```
 textview1.setOnClickListener(new View.OnClickListener() {
            @RequiresApi(api = Build.VERSION_CODES.M)
            @Override
            public void onClick(View view) {
                Intent iGallery = new Intent(Intent.ACTION_PICK) ;
                iGallery. setData
(MediaStore.Images.Media.EXTERNAL_CONTENT_URI) ;
                startActivityForResult(iGallery, GALLERY_REQ_CODE) ;

            }
        });
```

This code initializes two sequential preprocessing pipelines using TensorFlow's Keras API. The resize_and_rescale pipeline sequentially applies resizing and rescaling operations to input images. Initially, it resizes images to a specified size defined by IMAGE_SIZE. Subsequently, it normalizes pixel values by applying rescaling, dividing each pixel value by 255, which effectively scales them to the range [0,1]. On the other hand, the data_augmentation pipeline is designed for data augmentation, a technique used to enhance model generalization by introducing variations in training data. It applies two random transformations to input images: horizontal and vertical flips and random rotations with a maximum angle of 0.2 radians. These pipelines are fundamental in preprocessing and augmenting image data for tasks such as image classification, where they contribute to improving model performance and robustness.

```
resize_and_rescale = tf.keras.Sequential([
  layers.experimental.preprocessing.Resizing(IMAGE_SIZE, IMAGE_SIZE),
  layers.experimental.preprocessing.Rescaling(1./255),
])
data_augmentation = tf.keras.Sequential([
  layers.experimental.preprocessing.RandomFlip("horizontal_and_vertical"),
  layers.experimental.preprocessing.RandomRotation(0.2),
])
```

This Python function get dataset_partitions_tf facilitates the partitioning of a TensorFlow dataset (ds) into training, validation, and testing subsets based on specified split proportions. The function ensures that the sum of the train, validation, and test splits equals 1 to ensure a complete partitioning of the dataset. By default, it splits the dataset into 80% for training (train_split=0.8), 10% for validation (val_split=0.1), and 10% for testing (test_split=0.1). The dataset is optionally shuffled, and the shuffle size is set to 10000 by default.

```
def get_dataset_partitions_tf(ds, train_split=0.8, val_split=0.1,
test_split=0.1, shuffle=True, shuffle_size=10000):
    assert (train_split + test_split + val_split) == 1

    ds_size = len(ds)

    if shuffle:
        ds = ds.shuffle(shuffle_size, seed=12)

    train_size = int(train_split * ds_size)
    val_size = int(val_split * ds_size)

    train_ds = ds.take(train_size)
    val_ds = ds.skip(train_size).take(val_size)
    test_ds = ds.skip(train_size).skip(val_size)

    return train_ds, val_ds, test_ds
```

This code defines a convolutional neural network (CNN) model using TensorFlow's Keras API. The model architecture consists of several layers:

- **resize_and_rescale:** A preprocessing layer that resizes and rescales input images.
- **layers.Conv2D:** Convolutional layers with 32, 64, and 64 filters of size (3,3) respectively, followed by the ReLU activation function.
- **layers.MaxPooling2D:** Max-pooling layers with pool size (2,2) to downsample feature maps.
- **layers.Flatten:** Flattens the output of the previous layer into a 1D array.
- **layers.Dense:** Fully connected layers with 64 units and ReLU activation function.
- **layers.Dense:** Output layer with a softmax activation function, producing probabilities for each class.

The model architecture is suitable for image classification tasks, with input shape (BATCH_SIZE, IMAGE_SIZE, IMAGE_SIZE, CHANNELS) where BATCH_SIZE represents the number of samples in each batch, IMAGE_SIZE denotes the dimensions of input images, and CHANNELS

represents the number of color channels. The output layer has n_classes units corresponding to the number of classes in the classification task. Finally, the model is built with the specified input shape.

```
input_shape = (BATCH_SIZE, IMAGE_SIZE, IMAGE_SIZE, CHANNELS)
n_classes = 3

model = models.Sequential([
    resize_and_rescale,
    layers.Conv2D(32, kernel_size = (3,3), activation='relu',
input_shape=input_shape),
    layers.MaxPooling2D((2, 2)),
    layers.Conv2D(64,  kernel_size = (3,3), activation='relu'),
    layers.MaxPooling2D((2, 2)),
    layers.Conv2D(64,  kernel_size = (3,3), activation='relu'),
    layers.MaxPooling2D((2, 2)),
    layers.Conv2D(64, (3, 3), activation='relu'),
    layers.MaxPooling2D((2, 2)),
    layers.Conv2D(64, (3, 3), activation='relu'),
    layers.MaxPooling2D((2, 2)),
    layers.Conv2D(64, (3, 3), activation='relu'),
    layers.MaxPooling2D((2, 2)),
    layers.Flatten(),
    layers.Dense(64, activation='relu'),
    layers.Dense(n_classes, activation='softmax'),
])
model.build(input_shape=input_shape)
```

7.2 Results

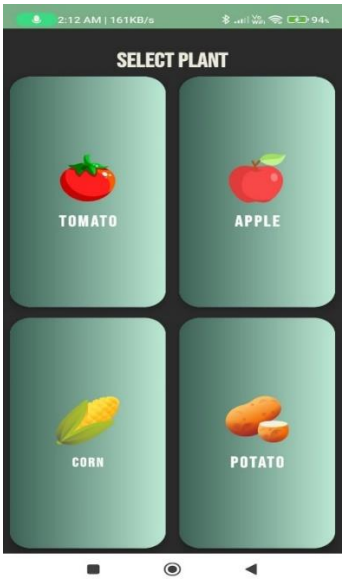

Fig 7.2.1: Main Screen of Application

The main screen of application shows that it can predict disease for tomato, apple, corn and potato.

Fig 7.2.2: Image of healthy Potato leaf

The above image shows that the uploaded image of the potato leaf is predicted to be healthy

Fig 7.2.3: Information about Tomato leaf disease

The application will provide the user information about the disease which has been affecting the crop leaf.

Crops	Model	Epoch	Accuracy
Potato	Sequential	50	0.95
	Inception	20	0.97
	Vgg16	20	0.98
Tomato	Sequential	10	0.76
	Inception	10	0.89
	Vgg16	10	0.95
Apple	Sequential	30	0.91
Corn	Mobilenetv2	75	0.92

Fig 7.2.4: CNN model and prediction accuracy

The above image depicts the different CNN models used for crops along with their accuracy and epochs.

CHAPTER 8

Project Plan

- **Gantt Chart**

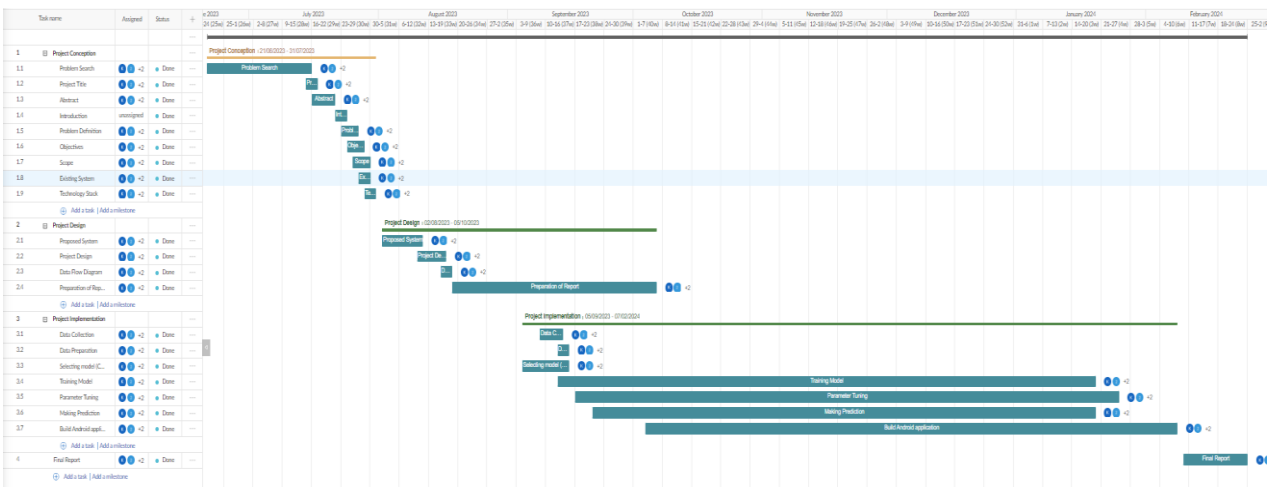

Figure 8.1 – Gantt Chart

CHAPTER 9

Conclusion

The expected outcome of Cropify android application is to effectively and accurately identify various diseases that are affecting potato, tomato, corn and apple by image feature extraction using leaves of the above plants. The application will be able to provide accurate analysis of the disease affecting the plant and in the future will be able to provide appropriate treatment for the plant disease to the farmers and gardeners. In future the application will be also be able to suggest information on preventive measures that can be implemented to avoid the spread of diseases. The application has been implemented using CNN sequential model to train and test the data that has been taken from the PlantVillage dataset to detect what disease plant is affected with by performing image feature extraction. In the android application the user will be able to capture the image of crop leaves using mobile phone camera or directly upload the image of the leaf from the mobile phone. The CNN model, then will analyze the image and detect which disease the leaf is affected with. Ultimately, the application will be user friendly, readily accessible which would make farmers and other users adopt it as a valuable tool for plant disease management. The crop of potatoes had the highest accuracy (98%), followed by that of tomatoes (95%), apples (91%), and corn (92%).

CHAPTER 10

Future Scope

Creating an Android application for predicting crop leaf diseases can have a significant future scope, especially as technology continues to advance and agriculture becomes more reliant on data-driven solutions. Here are some key areas of future growth and development for the application

1. **Integration with IoT and Sensor Technology:** The application can be enhanced by integrating with IoT devices and sensor technology. These sensors can collect real-time data from fields, such as temperature, humidity, soil moisture, and other relevant parameters. Integrating this data into the disease prediction model can improve the accuracy of predictions and provide farmers with timely alerts and recommendations.
2. **Machine Learning and AI Improvements:** As machine learning and AI algorithms continue to evolve, there will be opportunities to improve the accuracy and speed of disease prediction models. Implementing advanced machine learning techniques such as deep learning and ensemble models can enhance the application's ability to identify and diagnose crop diseases accurately.
3. **Database and Knowledge Base Expansion:** Continuously updating and expanding the application's database and knowledge base will be crucial for staying relevant and effective. Including information on new crop diseases, their symptoms, and management strategies will provide farmers with comprehensive and up-to-date resources.
4. **Collaboration with Agricultural Experts:** Collaborating with agronomists, plant pathologists, and other agricultural experts can enrich the application's content and ensure

the accuracy of disease identification and management recommendations. This collaboration can also help in validating the effectiveness of the application in real-world farming scenarios.

5. **Remote Sensing and Satellite Imagery:** Integrating remote sensing data and satellite imagery can provide a broader perspective on crop health across large agricultural areas. This data can be processed using image recognition techniques to detect early signs of diseases or stress in crops, allowing farmers to take proactive measures.
6. **User Interface and User Experience Enhancements:** Improving the user interface and experience of the application can enhance its usability and adoption among farmers. Intuitive design, easy navigation, multilingual support, and offline capabilities can make the application more accessible to users in diverse farming communities.
7. **Data Privacy and Security:** Ensuring robust data privacy and security measures will be critical to gaining trust among users. Implementing encryption protocols, secure data storage practices, and transparent data usage policies can protect farmers' sensitive information and build long-term credibility for the application.
8. **Market Expansion and Global Reach:** Expanding the reach of the application to different geographical regions and crops can broaden its impact and user base. Customizing the application to address specific crop diseases prevalent in different regions and providing localized content and recommendations can make it more valuable to farmers worldwide.

By focusing on these areas of development and leveraging emerging technologies, the application can have a promising future scope in revolutionizing agricultural practices and supporting sustainable crop management.

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Publication

The project entitled "Cropify: An Application for Plant Disease Detection" as submitted in the IEEE Conference named "2024 2nd World Conference on Communication & Computing 2024" and are waiting for acceptance status.

Cropify: An Application for plant disease detection

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***Abstract*— In order to guarantee global food security and economic stability, agriculture is essential. Crop diseases, however, pose a serious danger to crop productivity and quality and are just one of the many difficulties the agricultural industry must overcome. To lessen these dangers, early detection and prompt intervention are necessary. A cutting-edge crop disease prediction application for android to help farmers and agricultural stakeholders efficiently identify and manage crop diseases is developed.**

In order to deliver precise and immediate detection of crop illnesses, the Crop Disease Prediction Application makes use of the power of contemporary technology, including Deep learning and image recognition. With the help of the application, users can take pictures of infected crops on their Android handsets, which are then analysed using convolutional neural network. By analysing the photos and comparing them to a huge database of crop disease patterns, these algorithms enable quick diagnosis of the particular disease impacting the crop.

Keywords*— *CNN(Convolutional Neural Network), Deep Learning, Image Recognition, Disease Prediction

I. INTRODUCTION

In recent years, image processing and machine learning techniques have been used to develop automated crop disease detection systems. These systems can be used to rapidly and accurately detect and identify crop diseases from images of plant leaves, stems, and fruits. This project proposes to develop a crop disease prediction system using images of different classes of diseases. The system will be based on a Convolutional Neural Network model, which will be trained on a large dataset of labeled images of healthy and diseased crops. The system will first preprocess the input images to remove noise and improve the quality of the images. Then, it will extract features from the images that are relevant for disease detection. Finally, the system will use the extracted features to classify the images into healthy and diseased categories. The system will be evaluated on a held-out test set of labelled images. The evaluation metrics will include accuracy, precision, recall, and F1 score. The system is expected to be able to 2 accurately detect and identify a wide range of crop diseases. It is also expected to be robust to variations in lighting, background, and image quality. The project unfolds through several key phases, starting with the critical step of data collection and preprocessing. To train a robust and reliable deep learning model, an extensive and diverse dataset of labelled images is amassed, encompassing a wide spectrum of both healthy and diseased crops. The images of crops that have been used to train the deep learning model are potato and tomato. These images undergo meticulous preprocessing to eliminate noise and enhance image quality, ensuring that the model learns from pristine data. The subsequent phase entails feature extraction, in which convolutional neural networks (CNNs) take center stage. These networks extract pertinent features from the preprocessed images, allowing the model to discern and differentiate between healthy and diseased crops. This feature extraction process is vital in enabling the model to identify the unique characteristics and patterns that distinguish one class of disease from another. In conclusion, the Convolutional Neural Network based crop disease prediction project, with a focus on analyzing images of different classes of diseases, represents a remarkable convergence of cutting-edge technology and agricultural needs. By automating disease detection, this project addresses a longstanding challenge in agriculture, bridging the gap between traditional farming practices and the boundless possibilities offered by technological innovation. With its potential to enhance global food security through increased agricultural

productivity and reduced crop losses, this project stands as a testament to the transformative power of deep learning in the realm of agriculture.

II. RELATED WORK

The field of crop disease detection has attracted a lot of attention lately, which has prompted a number of studies aimed at utilizing cutting-edge technologies to counter the problem faced by farmers. Previous studies have demonstrated how well deep learning methods, image recognition and CNN can identify and predict the disease of the crop. Moreover research has shown using advanced CNN models to increase the accuracy of prediction of crop disease which adds to the increasing amount of research conducted regarding this topic.

Venkatesh et.al [1] concluded a convolutional neural network (CNN) model to categorise mango leaves infected with anthracnose. The fruit's yield and quality are impacted by the disease. The CNN model makes use of characteristics like leaf colour and shape and is based on the VGGNet and Inception models. At least 92% of the diseases could be accurately classified by the model uses the VGG16 model for feature extraction and fine-tuning, showing excellent classification accuracy for mango leaf disease. Jasim et.al [2] analysed deep learning algorithms to classify plant leaf diseases. The authors classified 15 types of disorders using a convolutional neural network (CNN), comprising 3 classes for healthy leaves and 12 classes for particular diseases. Images of potatoes, tomatoes, and peppers were among the data from the Plant Village dataset that were used in this investigation. For training and testing, the authors' accuracy rates were 98.29% and 98.029%, respectively. highlights how well image processing and deep learning work together to detect diseases in particular crops. Huang et.al [3] proposed the identification and categorization of grape leaf disease has been done using machine learning. It suggests four deep learning models that have been adjusted to perform better than pre-trained models, as well as an ensemble model that combines these four. The outcomes demonstrate the enormous potential of these models for grapevine production. Guan [4] analysed of a novel deep learning-based technique for plant disease detection. The technique achieves an 87% accuracy rate by combining four convolutional neural networks (CNNs) with a stacking technique. Comparing this to a single CNN model, there is a noticeable improvement. The scientists think that this technique could be applied as a cutting-edge plant disease warning system in real-world farming settings. Vallabhajosyula et.al [5] highlighted the performance of several pre-trained neural networks is thoroughly categorised in this research, along with the performance of a weighted ensemble of models pertinent to the identification of plant leaf diseases. Additionally, the plant village dataset—which consists of 38 classes gathered from 14 crops—is used to assess the performance of the proposed approach. The DENN model performs better than the most advanced pre-trained models, including ResNet 50 & 101, InceptionV3, DenseNet 121 & 201, MobileNetV3, and NasNet. The suggested model's performance evaluation shows that it performs comparably better than pre-trained models in classifying different plant illnesses. Siddiqui et.al [6] highlighted the use of CNN to forecast chilli leaf disease. Plant disease prognosis is not generally discussed. Sundari et.al [7] proposed the implementation of a deep convolutional network (CNN) plant disease recognition model for the early diagnosis and detection of plant diseases based on the categorization of leaf images. Swathi S et.al [8] analyzed convolutional neural networks for the identification of plant diseases. It provides an algorithm for precisely categorizing plant diseases and addresses how important it is to identify plant diseases in the agriculture sector. On the other hand, plant disease prediction with CNN is not mentioned in the paper specially. Ashritha et.al [9] concluded how deep learning, and more specifically augmented convolutional neural networks (CNNs), can be used to identify apple leaf illnesses in real time. Bhargav et.al [10] proposed Using CNN and Arduino to detect plant diseases can lead to early and precise diagnosis and combines low-cost microcontrollers with deep learning algorithms for real-time detection. Deshwal et.al [11] proposed a method for detecting plant leaf diseases that makes use of CNN and other machine learning techniques to provide accurate and effective plant disease diagnostics. Khan et.al [12] proposed a mobile-based system that specialises in maize crop diseases for the identification and categorization of plant leaf diseases. High prediction accuracy for disease detection was attained by utilising deep learning models. Ahmed et.al [13] analyzed the classification of palm leaf diseases is done using deep learning models (ResNet and Inception ResNet). Achieved 100% and 99.62% accuracy on both the original and supplemented datasets. Simhadri et.al [14] conducted the automatic detection of rice leaf diseases with the application of deep learning and transfer learning techniques. Choudhary et.al [15] concluded deep convolution neural network is effective in classifying leaf images for disease recognition.Accuracy, Precision, and Recall are used as evaluation parameters for performance assessment. Kasani et.al [16] proposed deep learning model used to identify and diagnose potato leaf diseases. Wei et.al [17] analyzed four distinct pre-trained CNN models were used in the creation and assessment of a deep learning model for the identification of plant leaf diseases. It contrasts these models' performances across various hardware and software setups.Convolutional Neural Network achieved 96.8% precision in classifying potato crop leaf. Kannagi et.al [18] highlighted a CNN model proposed for detecting plant diseases with high accuracy. The proposed CNN model achieves a greater accuracy of more than 95%. Aravindhan et.al [19] concluded the use of Convolutional Neural Networks (CNNs) for classifying crop diseases based on patterns recovered from diseased leaves. Ahmed et.al [20] proposed machine learning use to detect and classify plant diseases in leaves.Trained a CNN with 55,000 images and achieved 98.74% accuracy.

III. MATERIALS AND METHODS:

Dataset Description:

The dataset used in this study is the Plant Village dataset obtained from Kaggle, focusing on the plant species Solanum tuberosum (potato) and Solanum lycopersicum (tomato). The Plant Village dataset is a well-known collection of images showcasing various diseases affecting plant leaves. The dataset provides valuable resources for training and evaluating models for plant disease detection

Potato Dataset:

The subset of the Plant Village dataset dedicated to potatoes comprises a total of 2152 images. These images capture different stages of plant growth and various pathological conditions affecting potato leaves. The diseases covered include, but are not limited to, late blight, early blight, and healthy.

Tomato Dataset:

The tomato dataset within the Plant Village collection is more extensive, containing a total of 16,012 images. These images cover a wide range of tomato plant conditions, including healthy plants and those affected by diseases such as early blight, late blight, Bacterial spot, Leaf Mold, Septoria leaf spot, Spider mites Two spotted spider mite, Target Spot, Yellow Leaf Curl Virus and mosaic virus. The large number of images allows for a comprehensive exploration of the diverse manifestations of diseases in tomato plants.

Data Pre-processing:

Prior to model training, a series of pre-processing steps were applied to enhance the quality and uniformity of the dataset. These steps included resizing all images to a standardized resolution of 256x256 pixels. Additionally, pixel values were normalized to the range [0, 1] to ensure consistent input to the machine learning model. Augmentation techniques, such as random rotations and horizontal flips, were employed to augment the dataset, thereby improving the model's ability to generalize across different conditions. Data augmentation improves overfitting and increases validation accuracy [23].

Dataset Splitting:

To facilitate model training and evaluation, the dataset was split into three sets: 80% for training, 10% for validation and 10% for testing. The training set was used to optimize the model's parameters, while the validation set allowed for monitoring the model's performance on unseen data and early detection of overfitting.

This section provides an overview of the dataset used in the research, emphasizing the number of images for potatoes and tomatoes, along with the pre-processing steps applied to prepare the data for subsequent analysis and model training.

IV. PROPOSED METHODOLOGY

This section introduces a system designed to detect illnesses in plant leaves. There are multiple steps in the proposed system. The proposed system's main structure is depicted in Fig. 1.

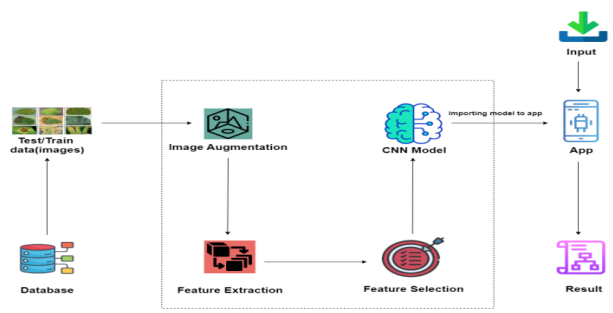

Fig.1:Architecture Diagram

A. *Data Acquistion:*

The main source of input data is the Plant Village Dataset, which consists of 16,012 photos for tomato and 21,52 images for potato that show different growth phases and disease situations.

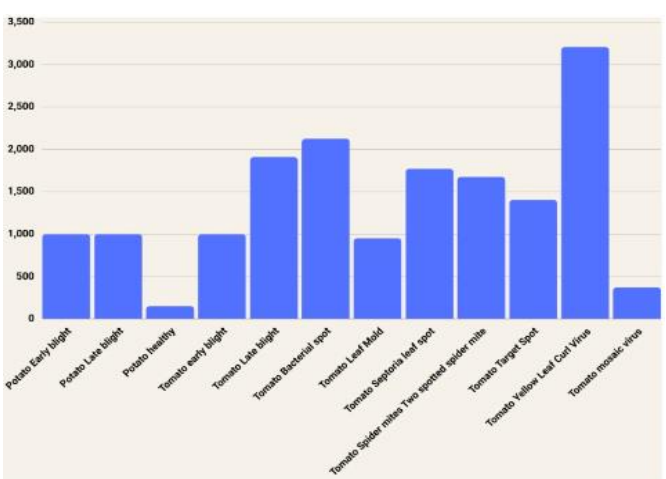

Fig.2:Different classes of crop diseases

B. *Image Pre-processing:*

The images in the dataset have been scaled to 256 by 256 resolutions in order to expedite the training process and

provide a realistic model testing computation. Train processing can be accelerated by optimising either the target or the input variables. while protecting the image's data integrity from loss. The dataset is split into three parts training, validation and testing. Dataset size is increased through augmentation to improve model generalisability.

C. *Feature Engineering:*

For crop disease detection, feature engineering transforms raw images into meaningful clues for the model. Think color and texture analysis, shape descriptors, and even advanced techniques like deep feature extraction. These features unveil the hidden whispers of disease within the pixels.

D. *Design of Convolutional Neural Network*

Comparative study of CNN-based disease detection with other traditional detection models using a potato plant leaf dataset [21]. One kind of deep neural network that is intended for processing structured grid data, like photographs, is the convolutional neural network (CNN). In computer vision applications including image classification, object recognition, and picture segmentation, CNNs have proven very popular and effective. A CNN's architecture usually comprises of many layers placed in a certain order. Let's dissect a CNN's main layers:

1. *Input Layer:*

In computer vision, the raw data is represented by the input layer, which is usually an image. Pixels are used to represent images, and each pixel in an RGB picture might contain many channels (red, green, and blue, for example).

2. *Convolutional Layer:*

Convolution is a mathematical procedure that is used by Convolutional Neural Networks (CNNs), which are feedforward neural networks with specialised layers for applying filters to input images [22]. CNNs are fundamentally composed of convolutional layers. Local feature extraction is accomplished by convolutional procedures, which entail swiping tiny filters—also referred to as kernels—over the input picture. Deeper layer patterns such as edges, textures, and higher-level features are recognised by these filters. Convolutional layers facilitate the capturing of feature spatial hierarchies.

3. *Activation Function (ReLU - Rectified Linear Unit):*

To add non-linearity, an activation function is applied element-by-element following the convolution process. ReLU is a commonly used activation function in CNNs, given by $f(x)=\max(0,x)$. ReLU helps the network learn complex patterns and accelerates convergence during training.

4. *Pooling Layer (Subsampling or Down-sampling):*

In CNN, pooling is accomplished by downsampling the feature maps by combining data from nearby regions. different pooling methods and their uses in computer vision, such as max pooling and average pooling [25]. The input volume's width and height are decreased by pooling layers. Max pooling and average pooling are two common pooling methods. Pooling contributes to the representation's increased invariance to slight input distortions and translations.

5. *Flatten Layer:*

Usually, a vector is obtained by flattening the spatial dimensions through many convolutional and pooling layers. Convolutional layers must be joined to completely connected layers using this flattening procedure.

6. *Fully Connected (Dense) Layers:*

Traditional neural network levels known as fully connected layers have every neuron in the layer above and below it connected to every other neuron. These layers combine the convolutional layers' high-level features and use the global context to inform their conclusions. The final prediction made by the network is often the result of the last completely linked layer. The performance of deep networks and learning techniques is strongly influenced by the selection of activation function.Neural networks and fully linked AI techniques can perform better when combined parametric activation functions are used [24].

7. *Output Layer:*

The ultimate categorization or prediction is generated by the output layer.The job at hand determines how many neurons are in the output layer (e.g., binary classification, multi-class classification).

8. *Softmax Activation (for Classification):*

The output layer in classification tasks frequently receives the application of the softmax activation function. To facilitate the interpretation of the model's predictions, Softmax transforms raw scores into probability distributions. Together, these layers make up a CNN's architecture. Through a technique known as

backpropagation, the network learns to modify its weights during training in order to maximise its predicting performance by minimising a certain loss function. Iterative training on labelled data is used throughout the process to refine the model until it performs well.

E. Training and Testing:

A Convolutional Neural Network (CNN) minimises a given loss function during training by using labelled data to update the model's weights using backpropagation. This is accomplished by using the training set. Furthermore, the validation set, which is a subset of the training set, aids in adjusting hyperparameters and avoiding overfitting. Following training, the CNN moves onto testing, when it assesses its performance with an additional batch of untested data. To make sure the model works well in real-world situations, performance measures like accuracy are calculated to evaluate the model's capacity to generalise to new cases.

F. Deployment:

A machine learning model is transformed into a mobile-friendly format (TensorFlow Lite for TensorFlow models) before being deployed into an Android app. The model is integrated into the Android app using the relevant libraries or frameworks after it has been optimised. The app's inference engine then applies the deployed model to additional data inputs to anticipate outcomes. Real-time processing limitations on mobile devices and memory efficiency are taken into account throughout this integration. Android Studio is also used to guarantee that the machine learning model running within the Android app performs at its best and to expedite the deployment process.

V. RESULTS

Crops	Model	Epoch	Accuracy
Potato	1.Sequential	50	0.95
	2.Inception	20	0.97
	3.Vgg16	20	0.98
Tomato	1.Sequential	10	0.76
	2.Inception	10	0.89
	3.Vgg16	10	0.95

Table.1:Comparative Study

The accuracy of various CNN models for tomato and potato crops is displayed in the table. The first column lists the crops, while the second column lists the models. The number of times the model has been trained using the data is known as the epoch. The percentage of accurate predictions the model produced is known as its accuracy.With 20 epochs and 97% accuracy, the Inception model produced the best results for potatoes. After 50 epochs, the Sequential model reached 95% accuracy, and after 20 epochs, the Vgg16 model reached 98% accuracy. The Vgg16 model had the best accuracy for tomatoes, with 95% accuracy after 10 epochs. After ten epochs, the Sequential model had achieved 76% accuracy and the Inception model had achieved 89% accuracy.Overall, on tomato and potato harvests, the Inception and Vgg16 models worked admirably. Although the Sequential model did not perform as well, it might be more computationally efficient.

Consider Sequential as an entry-level Lego set, Inception as a sophisticated, effective construction with sophisticated blocks, and VGG16 as a robust, albeit bulkier, castle. Sequential is easy to use and simple for beginners, Inception takes on more difficult tasks with greater accuracy but requires more resources, and VGG16 provides a good balance but performs worse than newer models. Select your weapon according to resource limitations, accuracy, and complexity.

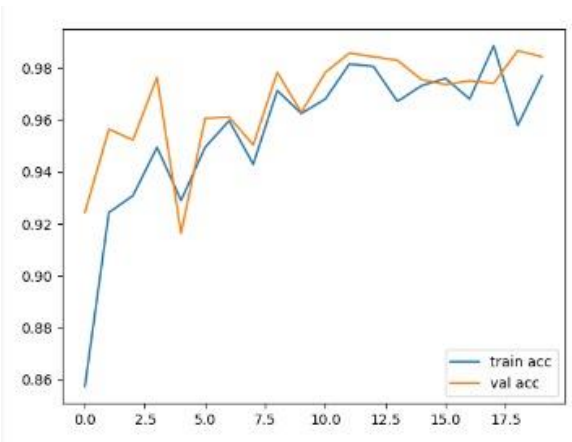

Fig.3:Training vs Validation Accuracy graph for potato vgg16 model

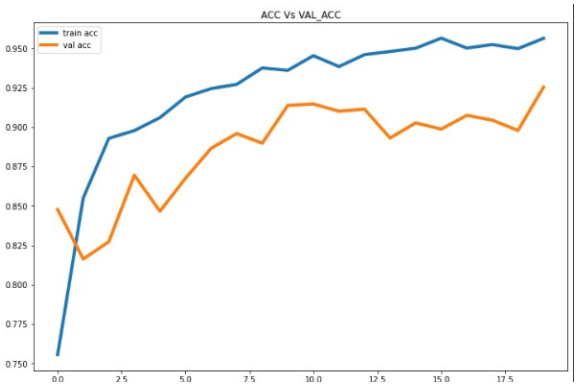

Fig.4:Training vs Validation Accuracy graph for tomato vgg16 model

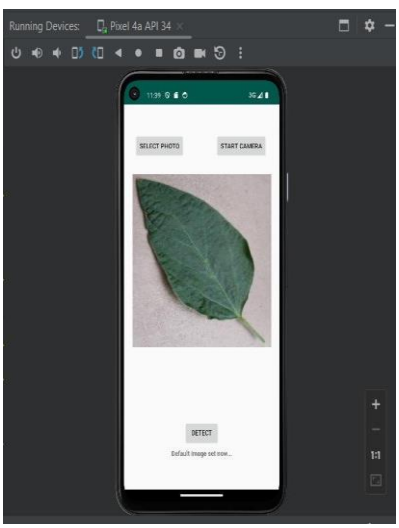

Fig.5: Uploading image on Cropify application

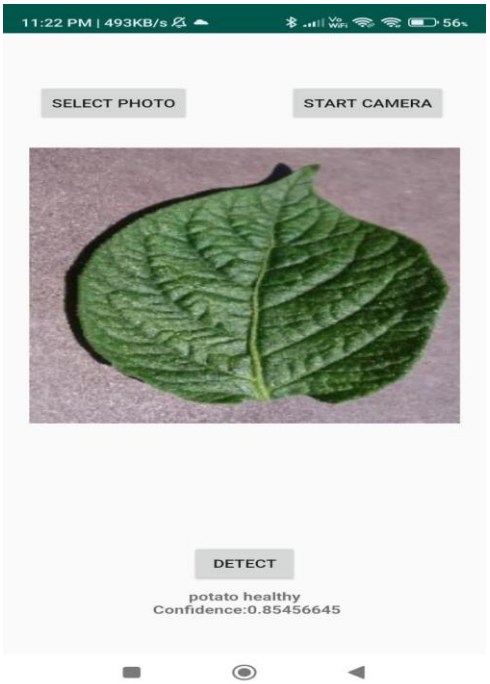

Fig.6: Detected image of healthy potato leaf

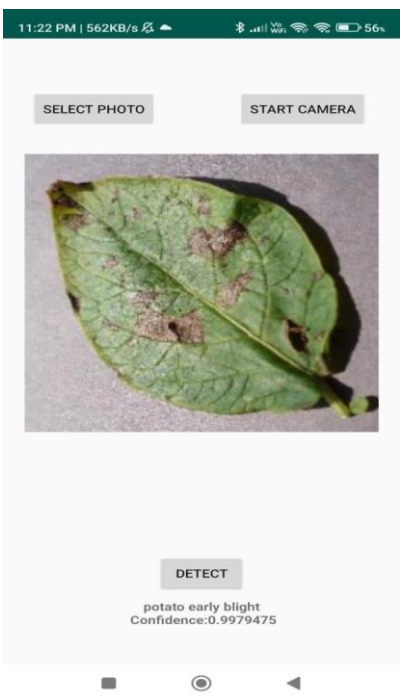

Fig.7:Detected image of potato leaf which has been affected with early blight

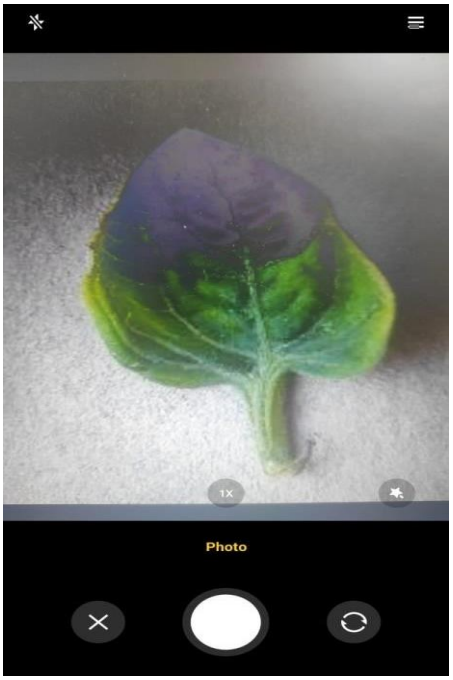

Fig.8:Taking a picture of tomato leaf

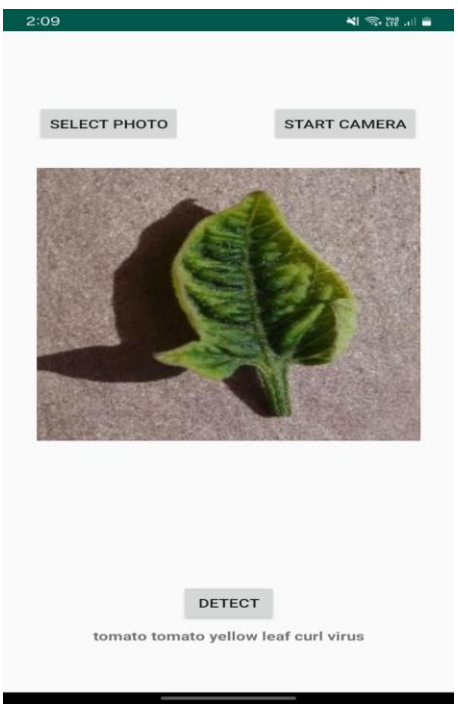

Fig.9: Detected image of tomato leaf which has been affected with yellow leaf curl virus

VI. CONCLUSION & FUTURE SCOPE

The goal of the Cropify Android application is to use picture feature extraction with the leaves of the aforementioned plants to effectively and precisely diagnose numerous diseases that affect tomatoes and potatoes. In the future, the application will be able to give farmers and gardeners the right therapy for plant diseases in addition to providing an accurate study of the disease impacting the plant. In the future, the app will also be able to recommend information on actions that may be taken to stop the spread of illness. The programme has been constructed with a CNN sequential model to perform image feature extraction and train and test the data acquired from the PlantVillage dataset in order to identify which disease plant is impacted. The Android application allows users to take pictures of potato or tomato leaves with their phones' cameras or to upload the photos straight from their phones. After that, the CNN model will examine the picture and determine whether disease is affecting the tomato or potato leaf. The application will ultimately be simple to use and easily available, which will encourage farmers and other users to embrace it as a useful tool for managing plant diseases

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